STUDYLAB HUB · COLLEGE CALCULUS I & AP CALCULUS · UNIT 1 (CHAPTER 1)

Unit 1: Limits, Continuity & Derivatives

One-Sided Limits · Algebraic Indeterminate Forms (0/0) · Squeeze Theorem · IVT · Limit Definition of the Derivative

College Calculus I & AP Calculus — Unit 1 (Chapter 1): Limits, Continuity & Definition of the Derivative

Independent, original study guide for College Calculus I & AP Calculus AB/BC (informed by OpenStax Calculus Volume 1, CC BY 4.0).

1. Unit 1 Roadmap (Chapter 1)

Section	Core Mathematical Definition	Key Exam Technique
1.1 · Understanding Limits	$\lim_{x \to c} f(x) = L$ if and only if the left-hand limit $\lim_{x \to c^-} f(x)$ and right-hand limit $\lim_{x \to c^+} f(x)$ both equal L	Check left-hand and right-hand limits separately on piecewise functions and absolute value expressions like $\text{abs}(x - a) \div (x - a)$
1.2 · Algebraic Limit Laws (0/0 Indeterminate Forms)	Factor & cancel, rationalize radicals using the conjugate, or clear common denominators	Never write "= 0/0" on a free-response exam (causes an immediate linkage-error deduction; write "limit takes the indeterminate form 0/0" instead)
1.3 · Trigonometric Limits & Squeeze Theorem	$\lim_{x \to 0} [\sin(kx) \div (kx)] = 1$ and $\lim_{x \to 0} [(1 - \cos x) \div x] = 0$	Bound oscillating products like $x^2 \cdot \sin(1/x)$ between $-x^2 \le x^2 \cdot \sin(1/x) \le x^2$
1.4 · Continuity at a Point (x = c)	Three-part test: (1) f(c) is defined, (2) $\lim_{x \to c} f(x)$ exists, and (3) $\lim_{x \to c} f(x) = f(c)$	Classify discontinuities as removable (hole), jump (LHL ≠ RHL), or infinite (vertical asymptote)
1.5 · Infinite Limits & Asymptotes	Vertical asymptote at x = a if $\lim_{x \to a\pm} f(x) = \pm\infty$; Horizontal asymptote y = L if $\lim_{x \to \pm\infty} f(x) = L$	When evaluating radical limits as $x \to -\infty$, remember that $\sqrt{(x^2)} = \text{abs}(x) = -x$ for negative x
1.6 · Intermediate Value Theorem (IVT)	If f(x) is **continuous** on [a, b] and k lies between f(a) and f(b), then there is at least one c in (a, b) where f(c) = k	Always state explicitly: *"Because f(x) is continuous on [a, b]..."* before applying IVT on FRQs
1.7 · Limit Definition of the Derivative	$f'(x) = \lim_{h \to 0} [f(x + h) - f(x)] \div h$ and $f'(a) = \lim_{x \to a} [f(x) - f(a)] \div (x - a)$	Recognize limits that are disguised derivatives, e.g. $\lim_{h \to 0} [\ln(e + h) - 1] \div h = d/dx[\ln x]$ at x = e, which equals 1/e

2. The 4 Essential Algebraic Limit Moves (0/0 Forms)

1. **Factoring & Canceling Common Factors**:

- Evaluate: **lim (x → 3) (x^2 – 9) ÷ (x^2 – 2x – 3)**
- Step 1: Factor numerator and denominator → lim (x → 3) [(x – 3)(x + 3)] ÷ [(x – 3)(x + 1)]
- Step 2: Cancel (x – 3) for x ≠ 3 and substitute x = 3 → (3 + 3) ÷ (3 + 1) = 6/4 = **3/2**

2. **Rationalizing Radicals with the Conjugate**:

- Evaluate: **lim (x → 0) [$\sqrt{}$(x + 4) – 2] ÷ x**
- Step 1: Multiply top and bottom by the conjugate **[$\sqrt{}$(x + 4) + 2]**:

Numerator becomes (x + 4) – 4 = x; Denominator becomes x · [$\sqrt{}$(x + 4) + 2]

- Step 2: Cancel x and substitute x = 0 → 1 ÷ ($\sqrt{}$4 + 2) = **1/4**

3. **Clearing Complex Fractions**:

- Evaluate: **lim (x → 2) [(1/x – 1/2) ÷ (x – 2)]**
- Step 1: Combine the numerator fractions over common denominator 2x → (2 – x) ÷ [2x(x – 2)]
- Step 2: Factor –(x – 2) in the numerator, cancel (x – 2), and substitute x = 2 → –1 ÷ (2 · 2) = **–1/4**

4. **Standard Trigonometric Limit Rewriting**:

- Evaluate: **lim (x → 0) sin(5x) ÷ sin(3x)**
- Step 1: Multiply and divide by 5x and 3x → lim (x → 0) [sin(5x) ÷ (5x)] × [(3x) ÷ sin(3x)] × (5/3)
- Step 2: Apply lim (u → 0) (sin u ÷ u) = 1 → 1 × 1 × (5/3) = **5/3**

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3. Worked Exam Problems

Problem 1 (Piecewise Continuity Parameter Solve)

Find the constants a and b that make the piecewise function f(x) continuous for all real numbers:

- **f(x) = 2x + 1** for x ≤ 1
- **f(x) = ax + b** for 1 < x < 3
- **f(x) = x^2 + 2** for x ≥ 3

Step-by-Step Solution:

1. **Continuity at x = 1**: Set LHL equal to RHL at x = 1:

- lim (x → 1^-) (2x + 1) = 2(1) + 1 = 3
- lim (x → 1^+) (ax + b) = a(1) + b = a + b
- Equation (1): **a + b = 3**

2. **Continuity at x = 3**: Set LHL equal to RHL at x = 3:

- lim (x → 3^-) (ax + b) = 3a + b
- lim (x → 3^+) (x^2 + 2) = 3^2 + 2 = 11
- Equation (2): **3a + b = 11**

3. **Solve the linear system**: Subtract Equation (1) from Equation (2):

- $(3a + b) - (a + b) = 11 - 3 \Rightarrow 2a = 8 \Rightarrow$ **$a = 4$**, and $b = 3 - 4 =$ **-1**.

Problem 2 (Horizontal Asymptotes with Radicals as $x \to -\infty$)

Evaluate: $\lim_{x \to -\infty} (3x - 2) \div \sqrt{(4x^2 + 5)}$

- **Step-by-Step Solution**:

1. Factor x^2 inside the square root: $\sqrt{(4x^2 + 5)} = \sqrt{(x^2)} \cdot \sqrt{(4 + 5/x^2)} = |x| \cdot \sqrt{(4 + 5/x^2)}$.
2. Because $x \to -\infty$ (so x is negative), **$|x| = -x$**.
3. Divide numerator and denominator by x:

$\lim_{x \to -\infty} (3 - 2/x) \div [-\sqrt{(4 + 5/x^2)}] = (3 - 0) \div (-\sqrt{4}) =$ **$-3/2$**.